

SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)
Department of Computer Science and Engineering
ASSESSMENT TOOL 2 – SCENARIO BASED ASSESSMENT (2 Questions)

Course Code:	ITA05	Course Title:	COMPUTER VISION
CO Assessed:	CO2 – Manipulation (BL3)	Assessment:	Assessment Tool 2 – Scenario Based Assessment
Weightage:	20% of CIA	Total Marks:	20 (2 Questions × 10 Marks)
CO5 Statement:	<i>Apply image enhancement and segmentation techniques for extracting meaningful regions from images.(BL3)</i>		
Student Name:		Reg. No.:	
Section / Batch:		Date:	

Code of Conduct

I _____ (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: _____

Instructions

- This test contains 2 questions, each carrying 10 marks. Total: 20 Marks.
- No negative marking. Unanswered questions carry zero marks.
- No calculators or electronic devices permitted.
- Duration: 60 minutes.

Section / Topic	Focus Area	Q. Nos.	Marks
Frequency Domain Image Enhancement	Periodic noise removal, Fourier Transform, notch filtering, quality inspection	1	10
Spatial Filtering & Edge-Preserving Enhancement	Bilateral filtering, edge preservation, autonomous vehicle navigation	2	10
GRAND TOTAL:			20 Marks

Annexure A – Scenario Based Assessment

Scenario 1: Manufacturing Quality Inspection

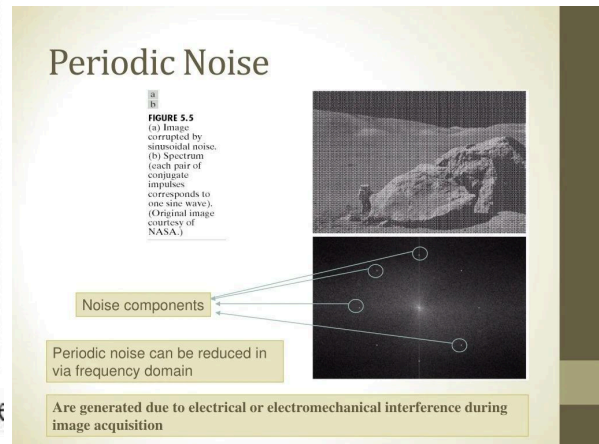
(a) Interpret the Scenario and Identify the Type of Noise



(a) Original Image



(b) Image with Periodic Noise



Interpretation

In an automated manufacturing industry, cameras capture images of metal components to detect defects such as:

- Surface cracks
- Scratches
- Holes
- Dents
- Missing parts

However, due to electrical interference from nearby machines, power lines, or electronic equipment, repetitive horizontal stripes appear across the captured images.

Type of Noise

The noise present is:

Periodic Noise (Sinusoidal Noise)

Characteristics:

- Appears as repetitive horizontal or vertical lines.
- Caused by electrical or electromagnetic interference.
- Exists at specific frequencies.
- Best handled in the frequency domain.

(b) Explain How This Degradation Affects Inspection Accuracy

The periodic stripes cover important image details.

Effects include:

- Small cracks become invisible.
- Surface scratches cannot be identified clearly.
- Object boundaries become distorted.
- Edge detection algorithms produce incorrect results.
- Machine learning models receive poor-quality images.
- False positives increase.
- False negatives increase.
- Overall inspection accuracy decreases.

Example

Without noise:

✓ Tiny crack detected.

With periodic noise:

✗ Crack hidden behind horizontal stripes.

Therefore, defective products may be incorrectly classified as good products.



(a)



(b)

(c) Recommend an Appropriate Enhancement Technique

Recommended Technique

Frequency Domain Notch Filter

Process:

1. Convert image to frequency domain using Fourier Transform (FFT).
2. Locate periodic noise frequencies.
3. Apply Notch Filter to remove only those frequencies.
4. Perform Inverse Fourier Transform.
5. Obtain a clean image.

(d) Justify Why Frequency Domain is Better than Spatial Domain

Frequency Domain Advantages

Frequency Domain

Removes only unwanted frequencies

Preserves fine details

Highly effective for periodic noise

Spatial Domain

Removes entire image regions

May blur important details

Poor for periodic noise

Maintains edge information

Can damage edges

Higher inspection accuracy

Lower inspection accuracy

Periodic noise appears as distinct peaks in the frequency spectrum, making it easy to isolate and remove.

Spatial filters such as Mean Filter or Median Filter cannot selectively remove periodic interference because they operate directly on neighboring pixels.

(e) Logical Sequence

Metal Component



Camera Captures Image



Electrical Interference



Periodic Horizontal Noise



Fourier Transform (FFT)



Locate Noise Frequencies



Apply Notch Filter



Inverse FFT



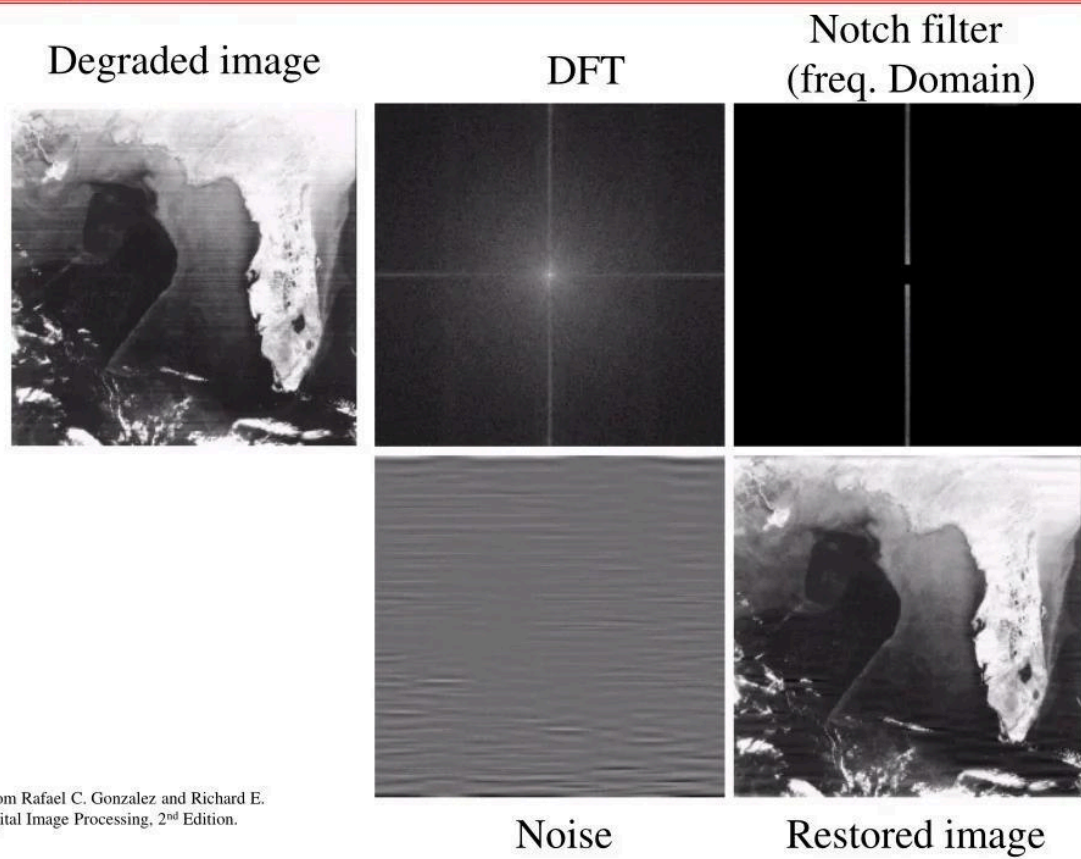
Clean Image



Accurate Defect Detection

Diagram of Frequency Domain Process

Notch Reject Filter:



(Images from Rafael C. Gonzalez and Richard E. Wood, Digital Image Processing, 2nd Edition.

Basic steps for filtering in the frequency domain

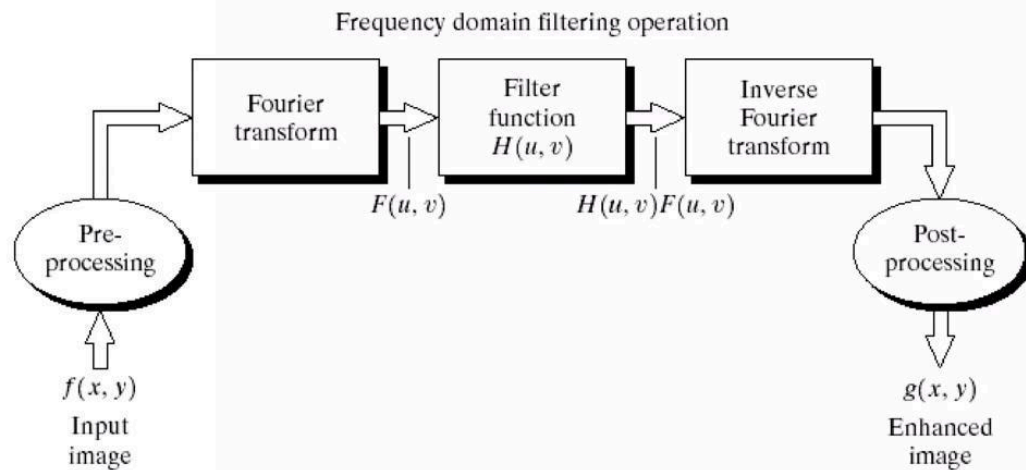


FIGURE 4.5 Basic steps for filtering in the frequency domain.

Scenario 2: Autonomous Vehicle Navigation

(a) Interpret the Scenario and Explain the Challenge

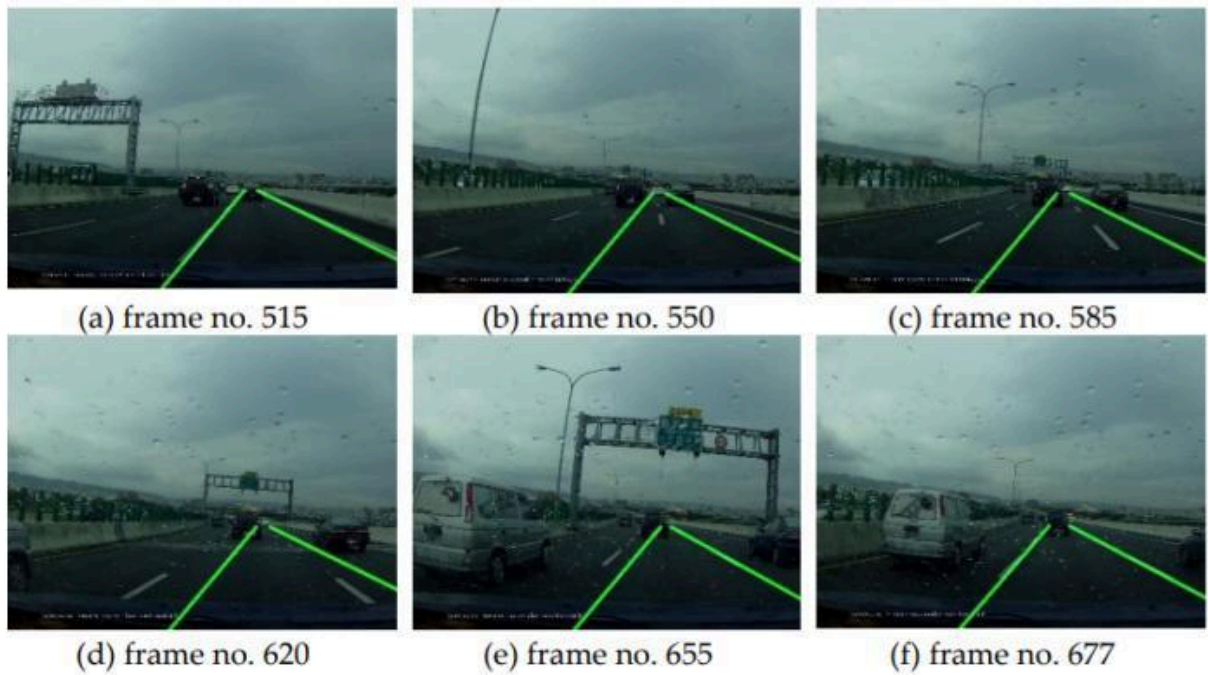


Figure 7. Automatic lane tracking while lane changing



(a) Input Image



(b) Noise Removed Image



(c) Contrast-Enhanced Image

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Interpretation

A self-driving car continuously captures road images to identify:

- Lane markings
- Road boundaries
- Traffic signs
- Vehicles
- Pedestrians

During heavy rain:

- Water droplets create image noise.
- Visibility decreases.
- Camera sensors introduce random noise.

The challenge is to remove noise while preserving important road features.

(b) Importance of Preserving Edges

Edges represent:

- Lane boundaries
- Road edges
- Vehicles
- Traffic signs
- Pedestrians

If edges become blurred:

- Lane detection fails.
- Object detection becomes inaccurate.
- Road boundary estimation becomes incorrect.
- Vehicle navigation becomes unsafe.

Therefore,

Noise should be removed without affecting edges.

(c) Recommend a Suitable Filtering Technique

Recommended Filter

Bilateral Filter

The Bilateral Filter considers:

- Spatial distance between pixels
- Pixel intensity similarity

It smooths homogeneous regions while preserving sharp edges.

(d) Justify Why Bilateral Filter Preserves Important Details

Advantages:

- Removes Gaussian-like image noise.
- Preserves lane markings.
- Maintains road edges.
- Keeps traffic signs sharp.
- Avoids excessive image blurring.
- Produces better edge preservation than Gaussian or Mean filters.

Comparison

Filter	Noise Removal	Edge Preservation
Mean Filter	Good	Poor
Gaussian Filter	Good	Moderate
Median Filter	Good for Salt & Pepper	Moderate
Bilateral Filter	Excellent	Excellent

The Bilateral Filter combines spatial closeness and intensity similarity, preventing smoothing across strong edges.

(e) Logical Sequence

Road Image



Heavy Rain



Random Noise



Apply Bilateral Filter



Noise Removed



Lane Markings Preserved



Road Edges Preserved



Safe Autonomous Navigation



Bilateral Filtering Illustration



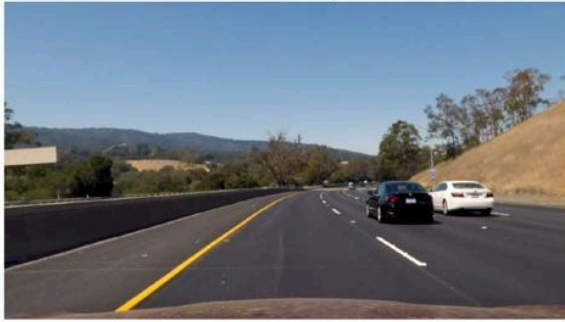
Bilateral Filter Example



Gaussian filter

Bilateral filter

1. Undistorted, Blurred



2. Color, Sobel Threshold



3. Warped to Top-Down



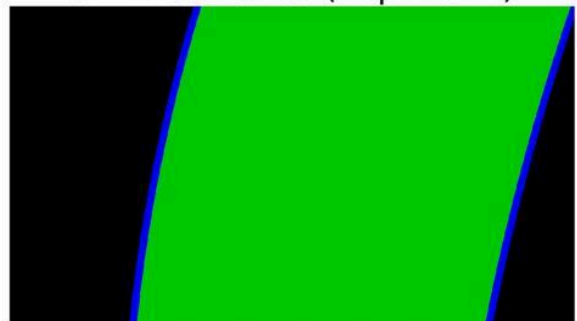
4. Masks for Left / Right Lanes



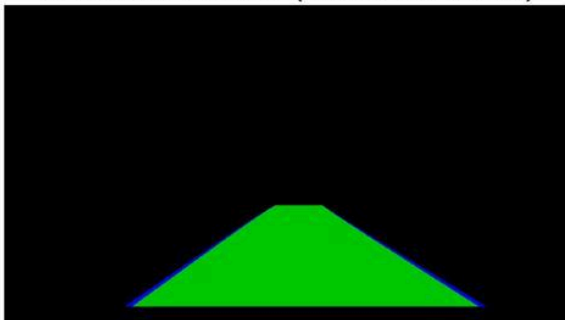
5. Masked Points



6. Fitted Lanes (Top Down)



7. Fitted Lanes (Camera View)



8. Final Result



Final Comparison (Exam Summary)

Feature	Scenario 1	Scenario 2
Application	Manufacturing Quality Inspection	Autonomous Vehicle Navigation
Problem	Horizontal stripe interference	Rain-induced image noise
Noise Type	Periodic (Sinusoidal) Noise	Random/Gaussian-like Rain Sensor Noise
Goal	Remove periodic interference	Remove noise while preserving edges
Enhancement Technique	Notch Filter (Frequency Domain)	Bilateral Filter
Domain Used	Frequency Domain	Spatial Domain (Edge-Preserving)
Main Advantage	Removes specific frequency components without affecting image details	Smooths noise while maintaining sharp lane markings and road boundaries
Outcome	Accurate defect detection and improved inspection quality	Reliable lane detection and safer autonomous driving